THE SPEED OF CONVERGENCE OF ONE ITERATIVE PROCESS*

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When solving linear elliptic equations by the method of finite differences it becomes a matter of solving systems of equations of the type

$$A\mathbf{u} = \mathbf{f};$$

where the order of the matrix A may be extremely high, so that exact methods of solution are unsuitable. Iterative methods of the type

$$\mathbf{u}^{\nu+1} = \mathbf{u}^{\nu} - \alpha\,(A\mathbf{u}^{\nu} - \mathbf{f})$$

are very slowly convergent when A is of high order, so that any methods for accelerating the convergence become very valuable. We shall consider here the simple equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f$$

in the square $0 \leqslant x \leqslant \pi;\ 0 \leqslant y \leqslant \pi$ with, for simplicity, zero boundary conditions.

We introduce the net $x_i = ih;\ y_j = jh;\ i, j = 0, 1, \ldots, N$ and define the difference operators as

$$\begin{aligned}(\Delta_1 u)_{i,j} &= u_{i-1,j} - 2u_{i,j} + u_{i+1,j},\\ (\Delta_2 u)_{i,j} &= u_{i,j-1} - 2u_{i,j} + u_{i,j+1}.\end{aligned} \qquad (i, j = 1, 2, \ldots, N-1)$$

We turn to the system of linear algebraic equations

$$\frac{1}{h^2}(\Delta_1 u)_{i,j} + \frac{1}{h^2}(\Delta_2 u)_{i,j} = f_{i,j},$$

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or

$$\Delta u \equiv (\Delta_1 + \Delta_2)\, u = h^2 f. \tag{1}$$

The quality of the iterative method of solution can be estimated as follows: we determine the norm of the "discrepancy" of the approximate solution:

$$\delta^\nu = \| \Delta u - h^2 f \| = \left[\sum_{i,j=1}^{N-1} h^2 (\Delta u - h^2 f)_{i,j}^2 \right]^{1/2},$$

where δ^ν is a magnitude of order $e^{-\gamma N^\nu}$. The quantity γ_N in fact characterizes the quality of the iterative process. For instance, we have $\gamma_N \sim 1/N^2$ for the elementary iterative process $u^{\nu+1} = u^\nu + \alpha (Au^\nu - fh^2)$ with certain restrictions on α. The iterative process $u^{\nu+1} = u^\nu + \alpha_\nu (\Delta u^\nu - h^2 f)$ described in [1] enables us to obtain $\gamma_N \sim 1/N$ with a special choice of the sequence α_ν.

A method with $\gamma_N \sim 1/\ln N$ was proposed in [2]. This method can be described schematically as follows:

$$u^{\nu+1/2} = u^\nu + \alpha_{\nu+1/2} (\Delta_1 u^{\nu+1/2} + \Delta_2 u^\nu - h^2 f),$$
$$u^{\nu+1} = u^{\nu+1/2} + \alpha_{\nu+1} (\Delta_1 u^{\nu+1/2} + \Delta_2 u^{\nu+1} - h^2 f);$$

α_ν is again a specially chosen sequence of numbers, which has an important influence on the quality of the method.

A relaxation method for solving such systems was proposed and tested in [3]. The same article discussed qualitatively the factors that influence the rate of convergence of this method. It must be remarked that, while the methods of [1] and [2] were based on commutative operators, so that estimation of the rate of convergence led to inequalities for a function on a spectrum (a polynomial in [1], a rational function in [2]), the method of [3] rests essentially on non-commutative operators, and estimations become extremely complex. Nevertheless, the calculations can be carried through to completion in a particular case, and there appear to be grounds for expecting that a method can be obtained by these means with γ_N independent of N, i.e. $\gamma_N > \gamma > 0$.

We now proceed to obtain an inequality of this type in the case of equation (1). We shall assume for simplicity that $N = 2^n$. Let $\varphi^{(l,m)}$ denote the eigenfunctions of the operator $\Delta \equiv \Delta_1 + \Delta_2$:

$$\varphi_{i,j}^{(l,m)} = q_N \sin lx \sin my \,|_{x=ih;\, y=jh} \quad (l,\ m = 1, 2, \ldots, N-1),$$

where q_N is a normalizing factor.

The eigenvalues $\lambda_{l,m}$ may be computed thus:

$$\lambda_{l,m} = \lambda_l + \lambda_m = -4\sin^2\frac{lh}{2} - 4\sin^2\frac{mh}{2}.$$

The iterative process

$$\mathbf{u}^{\nu+1} = \mathbf{u}^{\nu} + \alpha(\Delta\mathbf{u}^{\nu} - h^2\mathbf{f}) \tag{2}$$

is convergent for $0 < \alpha < 1/4$. The eigenfunctions of the iterative operator are obviously the same as for the operator Δ; the eigenvalues $\mu_{l,m}$ are computed from the formula $\mu_{l,m} = 1 + \alpha\lambda_{l,m}$. Let us take e.g. $\alpha = 1/5$. Let U be a solution of system (1): $\mathbf{u}^{\nu} = \mathbf{U} + \mathbf{v}^{\nu}$; then $\mathbf{v}^{\nu+1} = \mathbf{v}^{\nu} + \alpha\Delta\mathbf{v}^{\nu}$. We expand the function $\Delta\mathbf{v}^{\nu}$ into a series in eigenfunctions:

$$\Delta\mathbf{v}^0 = \sum_{l,m=1}^{N-1} c_{l,m}\varphi^{(l,m)}.$$

On performing ν_0 iterations in accordance with (2), we obtain

$$\Delta\mathbf{v}^{\nu_0} = \sum_{l,m=1}^{N-1} (\mu_{l,m})^{\nu_0} c_{l,m}\varphi^{(l,m)}. \tag{3}$$

The number of operations required for this is $A\nu_0 N^2$; here A is a constant which is independent of N.

We shall describe those eigenfunctions for which $l < N'$ and $m < N'$, $N' = 1/2\ N$, as "good", and the rest as "bad". Obviously, for the "bad" eigenfunctions, $|\mu_{l,m}| \leqslant q < 1$, where q is independent of N. In our case $q \sim 0.6$ when $\alpha = 1/5$. Thus, after a few iterations in accordance with (2) and (3), only the terms corresponding to "good" eigenfunctions will play a part. The main difficulty of the problem in fact lies in liquidating these. We shall use the following device for liquidating the "good" components of the error. We introduce the auxiliary net $x_i = i2h$, $y_j = j2h$.

We consider the function

$$\mathbf{V}_{2i,2j} = (\Delta\mathbf{v}^{\nu_0})_{2i,2j} \quad (i, j = 1, 2, \ldots, N' - 1).$$

Let $\delta^0 = \|\Delta\mathbf{v}^0\|$. Now, $\Delta\mathbf{v}^{\nu_0} = \Delta\mathbf{v}_1^{\nu_0} + \Delta\mathbf{v}_2^{\nu_0}$ represents a division into "good" and "bad" eigenfunctions, and $\|\Delta\mathbf{v}_2^{\nu_0}\| \leqslant q^{\nu_0}\delta^0$.

We introduce the operator

$$(\Delta_1' w)_{2i,2j} \equiv w_{2i-2,\,2j} - 2w_{2i,2j} + w_{2i+2,\,2j},$$

on the auxiliary net, together with the operators Δ'_2 and Δ'.

We shall now solve on the auxiliary net the equation (with zero boundary conditions)

$$(\Delta' w)_{2i,2j} = 4V_{2i,2j} + \|4v\|' \varepsilon_{2i,2j}. \tag{4}$$

Here $\varepsilon_{2i,2j}$ is an arbitrary function, subject only to the condition $\|\varepsilon_{2i,2j}\| < \varepsilon$; by introducing ε, we wish to emphasize that w is not too exact. The value of ε, which we shall determine later, is fixed.

We write w as $w = w_1 + w_2 + w_3$, with the assumption that $(\Delta' w_1)_{2i,2j} = 4(\Delta v_1)_{2i,2j}$, w_2 is connected with $\Delta v_2^{v_0}$, and w_3 is an inexact solution of equation (4).

We shall require the norm of the function on the auxiliary net; let us denote this by $\| \|'$:

$$\|\psi_{2i,2j}\|' = \left[\sum_{i,j=1}^{N'-1} 4h^2 (\psi_{2i,2j})^2\right]^{1/2}.$$

If a function $\psi_{i,j}$ is defined on the fundamental net, it is thereby also defined on the auxiliary net, and obviously

$$\|\psi\|' \leqslant 2\|\psi\|.$$

Hence $\|\Delta w_2\|' \leqslant 2q^{v_0}\delta^0$ and $\|\Delta w_3\|' \leqslant 4\varepsilon\|\Delta v^{v_0}\|' \leqslant 8\varepsilon\delta^0$.

The function $w_{2i,2j}$ is only defined on the "even" points of the fundamental net. We shall interpolate it linearly with respect to x and y on the remaining points of the fundamental net, i.e. we shall construct the function $W_{i,j}$ $(i, j = 0, 1, \ldots, N)$:

$$W_{i,j} = \begin{cases} w_{i,j}, \text{ if } i \text{ and } j \text{ are even;} \\ \frac{1}{2}(w_{i+1,j} + w_{i-1,j}), \text{ if } i \text{ is odd, } j \text{ is even;} \\ \frac{1}{2}(w_{i,j-1} + w_{i,j+1}), \text{ if } i \text{ is even, } j \text{ is odd;} \\ \frac{1}{4}(w_{i+1,j+1} + w_{i-1,j+1} + w_{i+1,j-1} + w_{i-1,j-1}), \text{ if } i \text{ and } j \text{ are odd.} \end{cases}$$

We now evaluate $\Delta_1 W$, $\Delta_2 W$ and ΔW. If i and j are even, then

$$(\Delta_1 W)_{i,j} = \tfrac{1}{2}(\Delta_1' w)_{i,j}, \quad (\Delta_2 W)_{i,j} = \tfrac{1}{2}(\Delta_2' w)_{i,j}, \quad (\Delta W)_{i,j} = \tfrac{1}{2}(\Delta' w)_{i,j}.$$

If i is even and j is odd, then

$$(\Delta_2 W)_{i,j} = 0,$$

$$(\Delta W)_{i,j} = (\Delta_1 W)_{i,j} = \tfrac{1}{4}(\Delta_1' w)_{i,j-1} + \tfrac{1}{4}(\Delta_1' w)_{i,j+1}.$$

If i is odd and j is even, then

$$(\Delta_1 W)_{i,j} = 0,\ (\Delta W)_{i,j} = (\Delta_2 W)_{i,j} = \tfrac{1}{4}(\Delta_2' w)_{i-1,j} + \tfrac{1}{4}(\Delta_2' w)_{i+1,j}.$$

If i and j are odd, then

$$(\Delta_1 W)_{i,j} = (\Delta_2 W)_{i,j} = (\Delta W)_{i,j} = 0.$$

We have thus established the following. Let the function w be defined only on the "even" points of the fundamental net, while W is obtained from w by linear interpolation at the remaining base-points of the net. Now, at each base-point, the function ΔW is some linear combination of the values of $\Delta_1' w$, $\Delta_2' w$ and $\Delta' w$ at the nearest "even" base-points. This enables us to use the familiar "triangle inequality" to write an inequality for $\|\Delta W\|$ in terms of $K\|\Delta' w\|$ (it follows from the fact that the operators Δ_1, Δ_2 are commutative and negative that $\|\Delta_1 w\| < \|\Delta w\|$). This simple but lengthy working is omitted, since all we need to know about K is that it is independent of $N (K < 1)$.

We can now evaluate the function $\tilde{v}^{\nu_0} = v^{\nu_0} - W$ on the fundamental net. We expand the function ΔW_1 into a series in eigenfunctions:

$$\begin{aligned} C_{l,m} = \sum_{i,j=1}^{N-1} h^2 (\Delta W_1)_{i,j} \varphi_{i,j}^{(l,m)} &= \tfrac{1}{2} h^2 \sum_{i,j=1}^{N'-1} (\Delta' w_1)_{2i,2j} \varphi_{2i,2j}^{(l,m)} + \\ &+ \tfrac{1}{4} h^2 \Big[\sum_{i,j=1}^{N'-1} (\Delta_1' w_1)_{2i,2j} \varphi_{2i,2j+1}^{(l,m)} + (\Delta_1' w_1)_{2i,2j} \varphi_{2i,2j-1}^{(l,m)} \Big] + \\ &+ \tfrac{1}{4} h^2 \Big[\sum_{i,j=1}^{N'-1} (\Delta_2' w_1)_{2i,2j} \varphi_{2i+1,2j}^{(l,m)} + (\Delta_2' w_1)_{2i,2j} \varphi_{2i-1,2j}^{(l,m)} \Big] = \\ &= h^2 \sum_{i,j=1}^{N'-1} (\Delta' w_1)_{2i,2j} \varphi_{2i,2j}^{(l,m)} + \tfrac{1}{4} h^2 \sum_{i,j=1}^{N'-1} (\Delta_1' w_1)\, (\Delta_2 \varphi^{(l,m)})_{2i,2j} + \\ &+ \tfrac{1}{4} h^2 \sum_{i,j=1}^{N'-1} (\Delta_2' w_1)\, (\Delta_1 \varphi^{(l,m)})_{2i,2j}. \end{aligned}$$

The only coefficients of interest will be those for which $l < N'$ and $m < N'$.

Notice that

$$(\Delta' w_1)_{2i,2j} = 4 \sum_{p,r=1}^{N'-1} (\mu_{r,p})^{\nu_0} c_{p,r} \varphi_{2i,2j}^{(p,r)}.$$

In the present case the eigenfunctions of the operator Δ' (denoted by $\Phi^{(p,r)}_{2i,2j}$ are connected with the $\varphi^{(p,r)}_{2i,2j}$ for $p < N'$ and $r < N'$ by the obvious relations

$$\Phi^{(p,r)}_{2i,2j} = \beta_N \varphi^{(p,r)}_{2i,2j},$$

where $\beta_N = 1 + O(1/N)$.

We shall assume N so large that the quantity $O(1/N)$ is less than some fixed number which we shall specify precisely later. Now,

$$C'_{l,m} = \sum_{i,j=1}^{N'-1} h^2 (\Delta' w_1)_{2i,2j} \varphi^{(l,m)}_{2i\,2j} = \frac{1}{4}(h')^2 \sum_{i,j=1}^{N'-1} (\Delta' w_1)_{2i,2j} \beta^{-1} \Phi^{(l,m)}_{2i,2j} =$$
$$= \beta^{-2} c_{l,m} (\mu_{l,m})^{\nu_0} = \left[1 + O\left(\frac{1}{N}\right)\right] (\mu_{l,m})^{\nu_0} c_{l,m}.$$

Using the fact that $(\Delta_1 \varphi^{(l,m)})_{2i,2j} = \lambda_l \varphi^{(l,m)}_{2i,2j} = \lambda_l \beta^{-1} \Phi^{(l,m)}_{2i,2j}$ и $(\Delta_2 \varphi^{(l,m)})_{2i,2j} = \lambda_m \beta^{-1} \Phi^{(l,m)}_{2i,2j}$, we finally obtain

$$C_{l,m} = \beta^{-2} c_{l,m} (\mu_{l,m})^{\nu_0} + \beta^{-1} \lambda_l C^1_{l,m} + \beta^{-1} \lambda_m C^2_{l,m}.$$

Here $C^1_{l,m}$ and $C^2_{l,m}$ are the coefficients of the expansions of the functions $\Delta'_1 w_1$ and $\Delta'_2 w_1$ (defined on the auxiliary net) into series in functions $\Phi^{(l,m)}$. Thus

$$\left[\sum_{l,m=1}^{N'-1} (C^1_{l,m})^2\right]^{1/2} = \| \Delta'_1 w_1 \|' \leqslant \| \Delta' w_1 \|',$$
$$\left[\sum_{l,m=1}^{N'-1} (C^2_{l,m})^2\right]^{1/2} \leqslant \| \Delta' w_1 \|'.$$

We have thus found upper bounds for the Fourier coefficients of the function $\Delta v_1^{\nu_0} - \Delta W_1$ for the "good" eigenfunctions. As regards its expansion in "bad" eigenfunctions, we only need the coarse inequality:

$$\| \Delta v_1^{\nu_0} - \Delta W_1 \| \leqslant \| \Delta v_1^{\nu_0} \| + \| \Delta W_1 \| \leqslant (1+K) \| \Delta v_1^{\nu_0} \| \leqslant (1+K)\delta^0.$$

We finally obtain:

$$\Delta \tilde{v}^{\nu_0} = -O\left(\frac{1}{N}\right) \sum_{l,m=1}^{N'-1} (\mu_{l,m})^{\nu_0} c_{l,m} \Phi^{(l,m)} -$$
$$- \beta^{-1} \sum_{l,m=1}^{N'-1} \lambda_l C^1_{l,m} \Phi^{(l,m)} - \beta^{-1} \sum_{l,m=1}^{N'-1} \lambda_m C^2_{l,m} \Phi^{(l,m)} +$$

$$+\Delta v_2^{\nu_0}-\Delta W_2-\Delta W_3+\sum_{l,\,m=N'}^{N-1} C_{l,m}\varphi^{(l,m)}.$$

The fourth, fifth, sixth and seventh terms on the right-hand side have the upper bounds $q^{\nu_0}\delta^0$, $2q^{\nu_0}\delta^0$, $8\varepsilon\delta^0$ and $(1+K)\delta^0$ respectively. As a result of ν_0 iterations in accordance with formula (2) and after using the auxiliary net, we have obtained the function $\tilde{v}_{i,j}^{\nu_0}$; we iterate this approximate solution ν_1 times in accordance with (2) and obtain the function $v_{i,j}^{\nu_0+\nu_1}$. We now consider $\|\Delta v^{\nu_0+\nu_1}\|$:

$$\|\Delta v^{\nu_0+\nu_1}\| \leqslant O\left(\frac{1}{N}\right)\delta^0+\beta^{-1}\max_{l,m}|\lambda_l(\mu_{l,m})^{\nu_1}|\,\|\Delta' w\|'+$$

$$+\beta^{-1}\max_{l,m}|\lambda_m(\mu_{l,m})^{\nu_1}|\,\|\Delta' w\|'+3q^{\nu_0}\delta^0+8\varepsilon\delta^0+4q^{\nu_1}\delta^0;$$

since $\|\Delta' w_1\| \leqslant 3\|\Delta v^0\| \leqslant 3\delta^0$, we need to consider

$$\max_{l,m}|\lambda_l(\mu_{l,m})^{\nu_1}| \leqslant \max|(\lambda_l+\lambda_m)(\mu_{l,m})^{\nu_1}|.$$

The function $f(\lambda)=\lambda(1-\alpha\lambda)^{\nu_1}$ (where $\alpha=\frac{1}{5}$, $0\leqslant\lambda\leqslant 8$) has a maximum at the point $f'(\lambda_{max})=0$:

$$f'(\lambda)=(1-\alpha\lambda)^{\nu_1-1}[1-\alpha\lambda(1+\nu_1)], \qquad \lambda_{max}=\frac{1}{\alpha(1+\nu_1)}.$$

Obviously $|f(\lambda_{max})| < |\lambda_{max}|$.

We now insert for definiteness $\varepsilon=\varepsilon_0=10^{-2}$ and $O(1/N)<\varepsilon/5$. We fix ν_1 in such a way that $3\beta^{-1}\lambda_{max}<\frac{1}{5}\varepsilon_0$ and $(1+K)q^{\nu_1}<\frac{1}{5}\varepsilon_0$; we take ν_0 so that $3q^{\nu_0}<\frac{1}{5}\varepsilon_0$, now obviously, $\|\Delta v^{\nu_0+\nu_1}\| \leqslant 9\varepsilon\delta^0 = 9\varepsilon\|\Delta v^0\|$.

Starting from the function $v^{\nu_0+\nu_1}$, we repeat the whole of the above cycle; now

$$\|(\Delta v^{\nu_0+\nu_1})^{\nu_0+\nu_1}\| \leqslant 9\varepsilon\|\Delta v^{\nu_0+\nu_1}\| \leqslant 9^2\varepsilon^2\|\Delta v^0\| \leqslant \varepsilon\|\Delta v^0\|.$$

We have fixed ε_0 so that $9^2\varepsilon_0^2 \leqslant \varepsilon_0$.

We shall now state our result.

In order to reduce the norm of the initial discrepancy by a factor ε_0^{-1} by the proposed method, we required $2(\nu_0+\nu_1)$ iterations in

accordance with formula (2), i.e. $2AN^2(\nu_0 + \nu_1)$ operations, and we had to solve twice the equation $\Delta' w = f$ up to the term ε_0* on the auxiliary net. If we take the 0-function as the initial function w^0 when solving the auxiliary equation, the initial value of the discrepancy for it will be $\|f\|'$, and, since we require the function w satisfying the equation $\Delta' w = f + \|f\|'\varepsilon$, we have arrived on the auxiliary net at the problem equivalent to that just solved on the fundamental net with the aid of the auxiliary net. In addition, we had twice to interpolate the function from the auxiliary to the fundamental net; this requires BN^2 operations. Let $Q(N, \varepsilon_0)$ now denote the number of operations needed in order to reduce the norm of the discrepancy by a factor ε_0^{-1} in the net $h = \pi/N$. In this case

$$Q(N, \varepsilon_0) = [2(\nu_0 + \nu_1)A + B]N^2 + 2Q\left(\frac{N}{2}, \varepsilon_0\right) = CN^2 + 2Q\left(\frac{N}{2}, \varepsilon_0\right) =$$
$$= CN^2 + 2\left[C\frac{N^2}{4} + 2Q\left(\frac{N}{4}, \varepsilon_0\right)\right] = \cdots = N^2C\left(1 + \frac{1}{2} + \frac{1}{4} + \cdots\right) \leqslant C_1N^2.$$

This is not completely strict, since we have the condition $O(1/N) < \varepsilon_0/5$, though this latter plays no part, i.e. it is not obligatory to solve Poisson's problem in each net with the aid of an auxiliary net (if $N/2^k < N_{min}$, we can solve the problem up to ε_0 by any method in a net with the interval $h = 2^k\pi/N$: it is obvious that this has no effect on the upper bound). The proposed method thus consists of a solution with the aid of an auxiliary net; if this latter is extremely large, the problem can also be solved on it by using a net of a particular type for the problem, and so on. It may easily be seen that the storage capacity required for realization of all auxiliary nets is not more than half the capacity required for storing the quantities $u^{\nu}_{i,j}$ and the right-hand sides of the equation $f_{i,j}$.

We have obtained an upper bound for the rate of convergence of the iterations with fairly special assumptions, which had to be made in order to facilitate otherwise complex working; our result is therefore not a complete proof, though it provides cogent grounds for believing that a good iterative method can be devised on the proposed lines. Practical experience has supported this (for a detailed description, see [3]). The practical form of the method may naturally differ from the form used to obtain the upper bounds, as was indeed the case in [3]. The choice of α in (2) requires specific information on the spectrum. In [3], therefore, instead of iterations in accordance with (2), Seidel

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* In the sense of formula (4).

iterations were employed, with which, as may easily be verified, satisfactory liquidation of the high-frequency components of the error is achieved.

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